

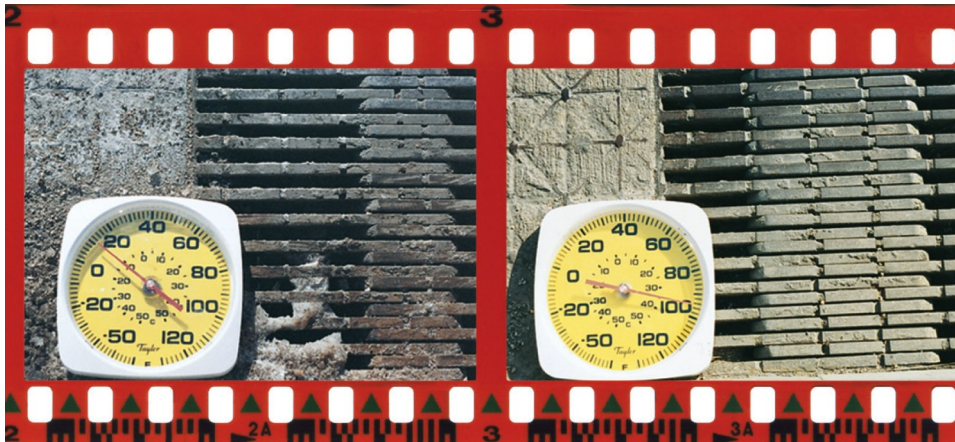
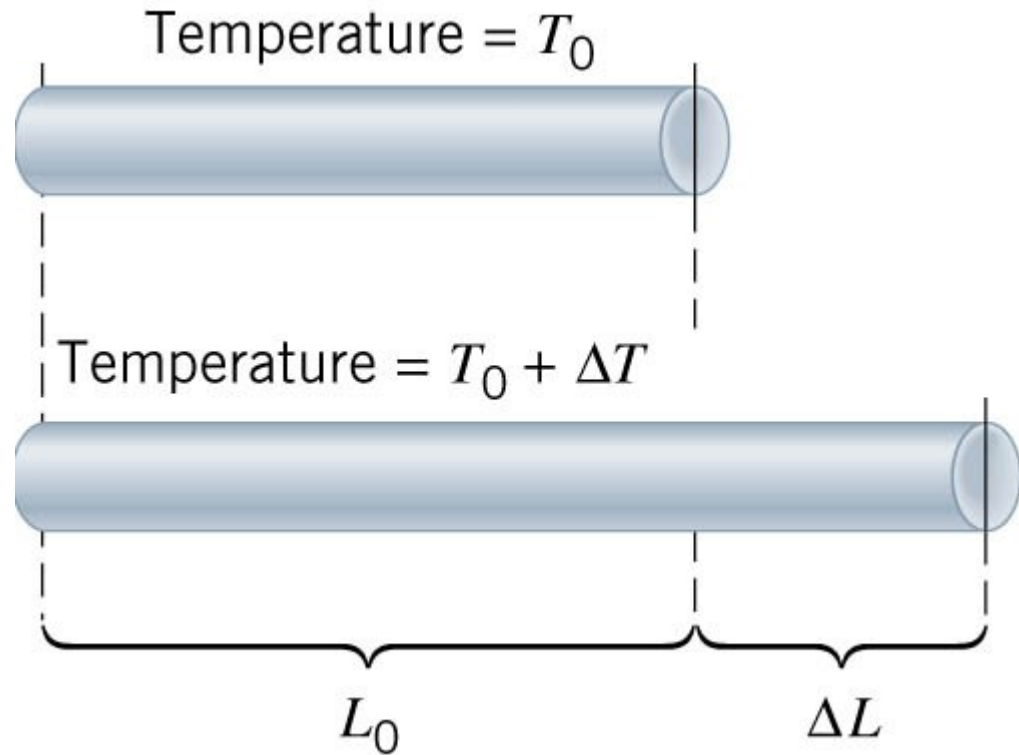
Thermal expansion

References: - Physics / Johnson and Cutnell / Wiley (slides)
- Conceptual Physics / Paul Hewitt / Pearson
- Physics for future presidents / Dr. Richard Muller / Princeton

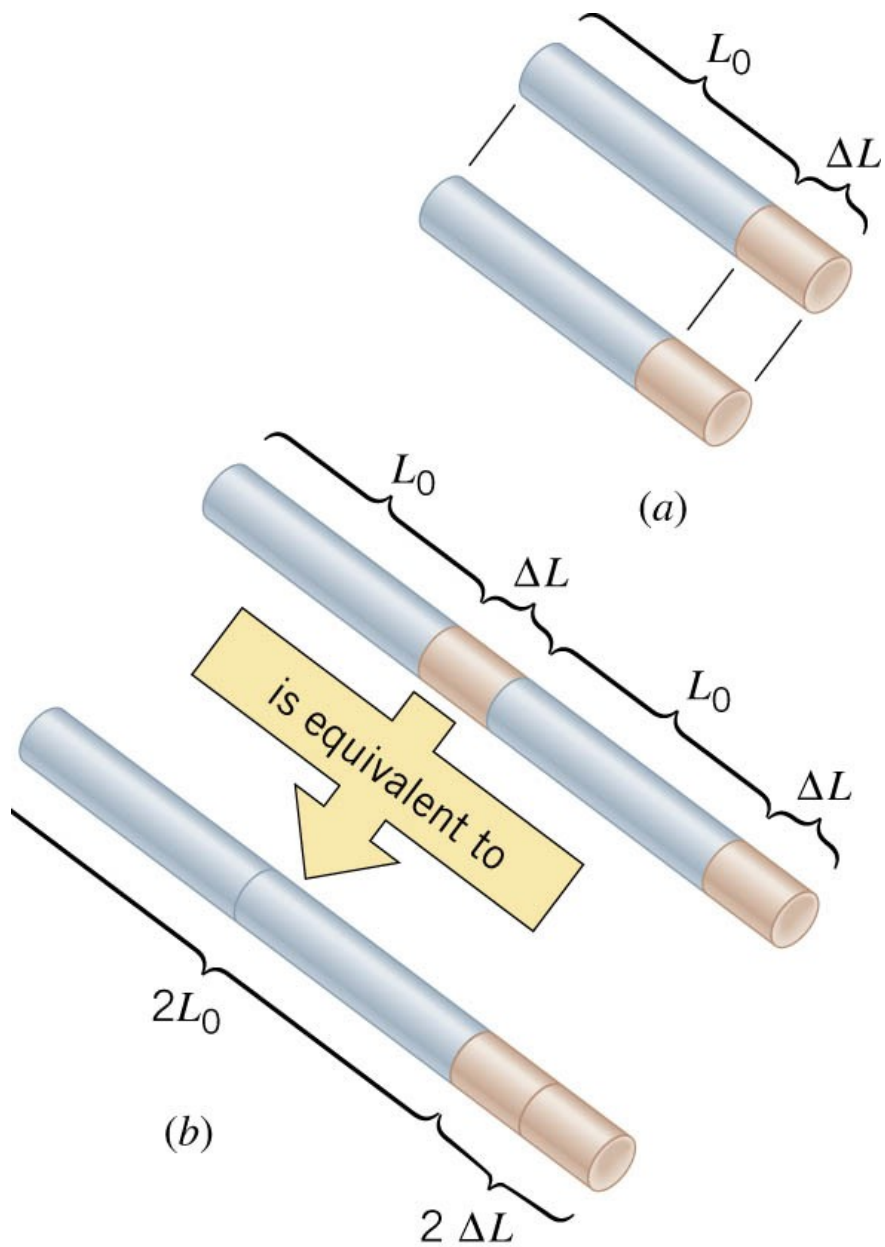


12.4 Linear Thermal Expansion

NORMAL SOLIDS



12.4 Linear Thermal Expansion

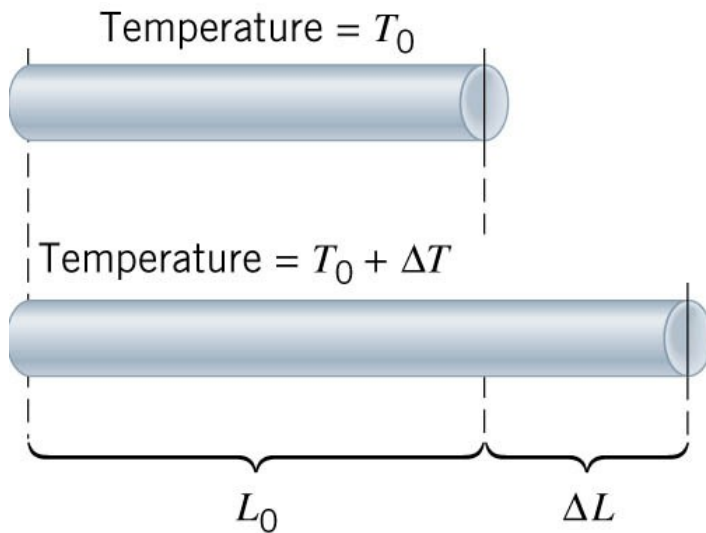


$$\Delta L \propto L_0$$

12.4 Linear Thermal Expansion

LINEAR THERMAL EXPANSION OF A SOLID

The length of an object changes when its temperature changes:



$$\Delta L = \alpha L_0 \Delta T$$

coefficient of
linear expansion

Common Unit for the Coefficient of Linear Expansion: $\frac{1}{C^\circ} = (C^\circ)^{-1}$

12.4 Linear Thermal Expansion

Table 12.1 Coefficients of Thermal Expansion for Solids and Liquids^a

Substance	Coefficient of Thermal Expansion (C°) ⁻¹	
	Linear (α)	Volume (β)
Solids		
Aluminum	23×10^{-6}	69×10^{-6}
Brass	19×10^{-6}	57×10^{-6}
Concrete	12×10^{-6}	36×10^{-6}
Copper	17×10^{-6}	51×10^{-6}
Glass (common)	8.5×10^{-6}	26×10^{-6}
Glass (Pyrex)	3.3×10^{-6}	9.9×10^{-6}
Gold	14×10^{-6}	42×10^{-6}
Iron or steel	12×10^{-6}	36×10^{-6}
Lead	29×10^{-6}	87×10^{-6}
Nickel	13×10^{-6}	39×10^{-6}
Quartz (fused)	0.50×10^{-6}	1.5×10^{-6}
Silver	19×10^{-6}	57×10^{-6}
Liquids^b		
Benzene	—	1240×10^{-6}
Carbon tetrachloride	—	1240×10^{-6}
Ethyl alcohol	—	1120×10^{-6}
Gasoline	—	950×10^{-6}
Mercury	—	182×10^{-6}
Methyl alcohol	—	1200×10^{-6}
Water	—	207×10^{-6}

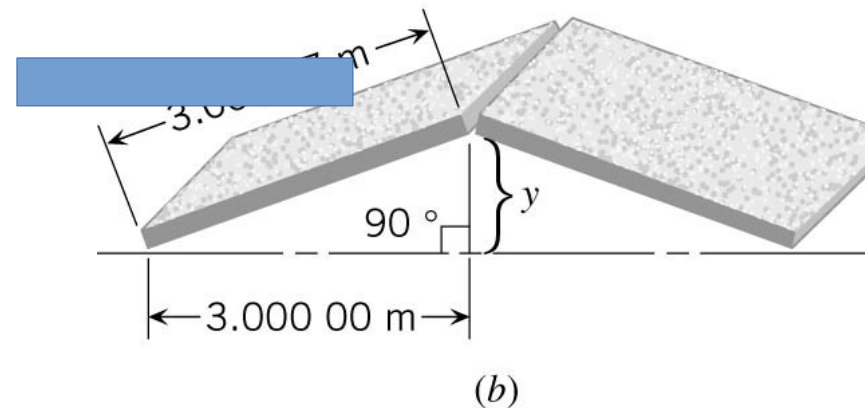
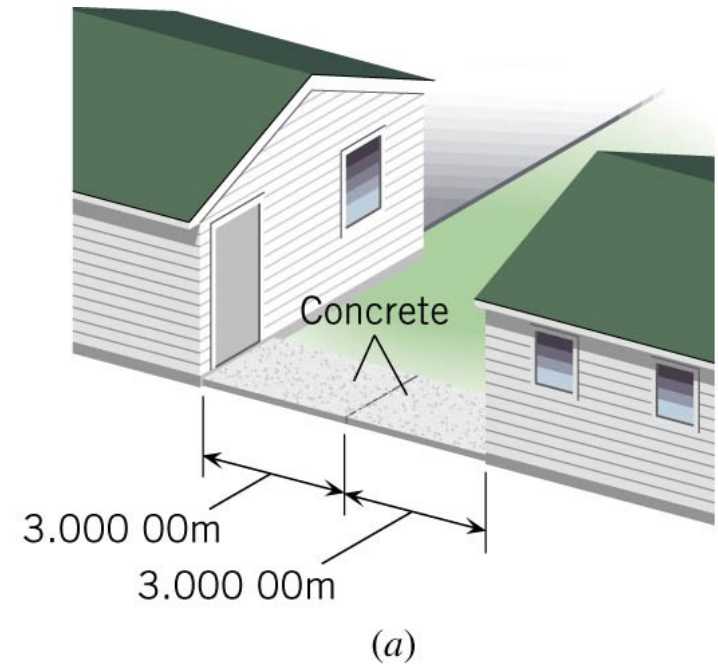
^aThe values for α and β pertain to a temperature near 20 °C.

^bSince liquids do not have fixed shapes, the coefficient of linear expansion is not defined for them.

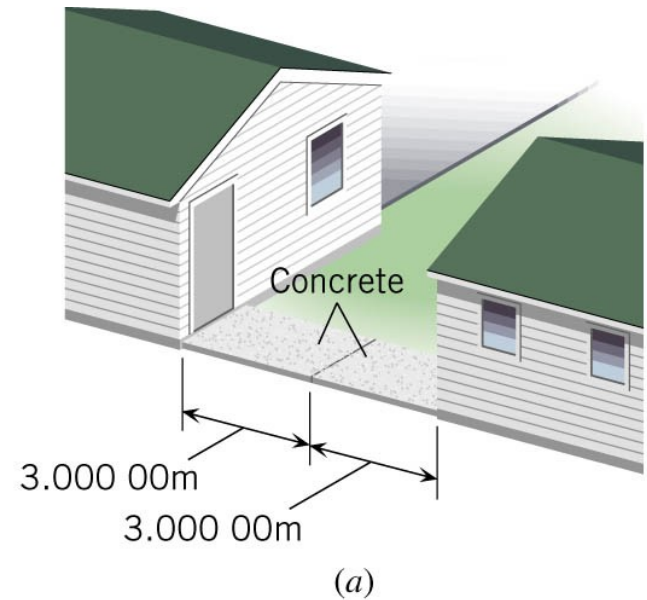
12.4 Linear Thermal Expansion

Example 3 The Buckling of a Sidewalk

A concrete sidewalk is constructed between two buildings on a day when the temperature is 25°C . As the temperature rises to 38°C , the slabs expand, but no space is provided for thermal expansion. Determine the distance y in part (b) of the drawing.

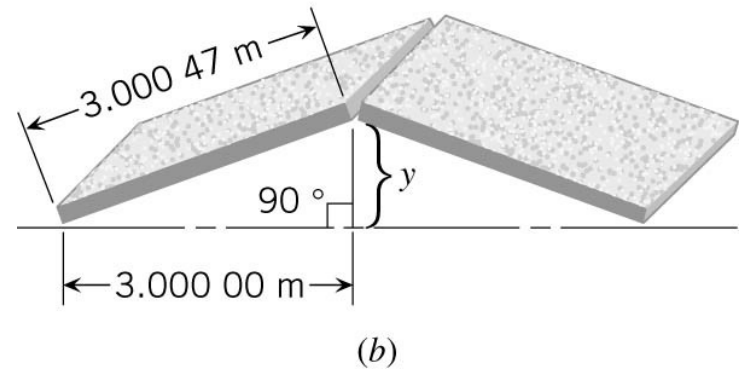


12.4 Linear Thermal Expansion



$$\Delta L = \alpha L_o \Delta T$$

$$\dot{\iota} \left[12 \times 10^{-6} (C^\circ)^{-1} \right] (3.0 \text{ m}) (13 \text{ C}^\circ) = 0.00047 \text{ m}$$



$$y = \sqrt{(3.00047 \text{ m})^2 - (3.00000 \text{ m})^2} = 0.053 \text{ m}$$

Losing Gas?

75°F day (24°C)

20 gallon steel gas tank

Fill with gas at 50°F (10°C)

How much gas would get lost?

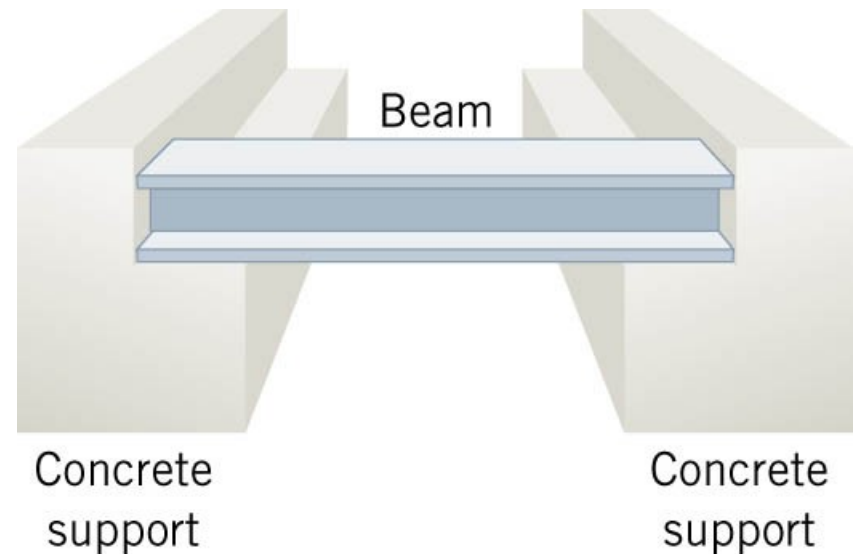
Material	Type	Expansion coefficient (°C ⁻¹)
Glass	Length	3.2×10^{-6}
Ice	Length	51×10^{-6}
Steel	Length	10×10^{-6}
Air	Volume	3.7×10^{-3}
Gasoline	Volume	95×10^{-5}
Water, 1°C	Volume	-5×10^{-5}
Water, 20°C	Volume	20×10^{-5}



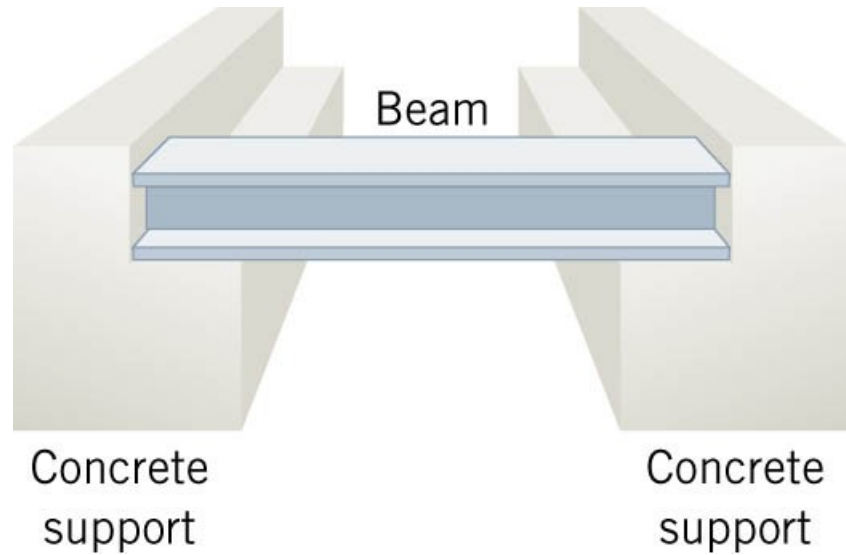
12.4 Linear Thermal Expansion

Example 4 The Stress on a Steel Beam

The beam is mounted between two concrete supports when the temperature is 23°C . What compressional stress must the concrete supports apply to each end of the beam, if they are to keep the beam from expanding when the temperature rises to 42°C ?



12.4 Linear Thermal Expansion



$$\Delta L = \alpha L_o \Delta T$$

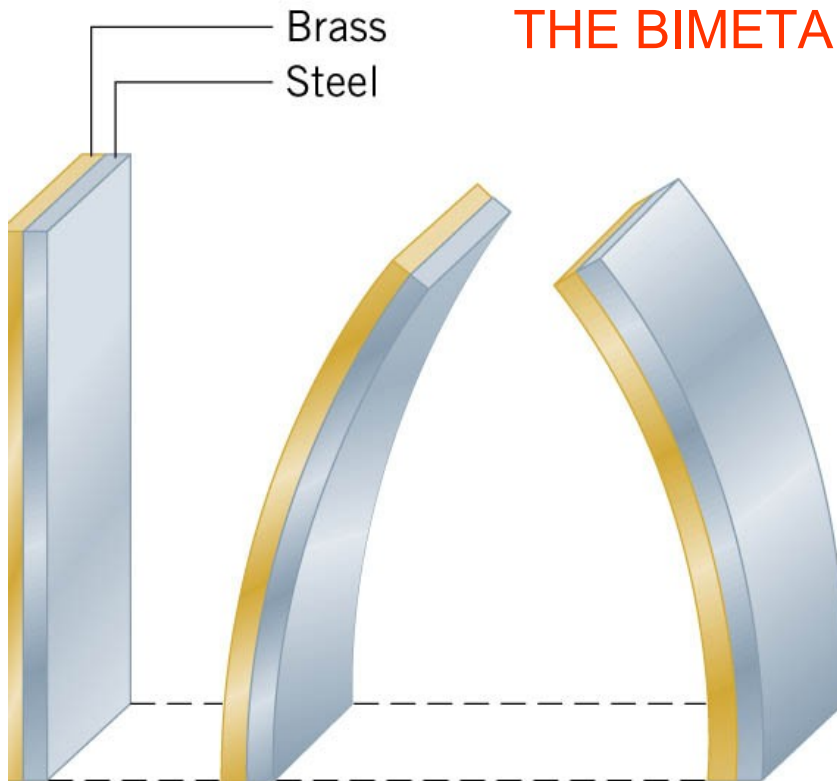
An orange curved arrow points from the ΔL term in the equation above to the ΔL term in the equation below.

$$\text{Stress} = \frac{F}{A} = Y \frac{\Delta L}{L_o} = Y \alpha \Delta T$$

$$(2.0 \times 10^{11} \text{ N/m}^2) [12 \times 10^{-6} (\text{C}^\circ)^{-1}] (19 \text{ C}^\circ) = 4.7 \times 10^7 \text{ N/m}^2$$

12.4 Linear Thermal Expansion

THE BIMETALLIC STRIP

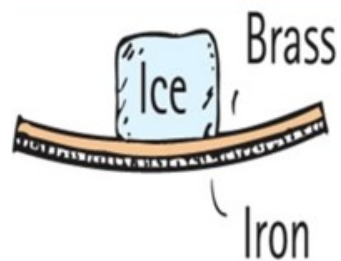
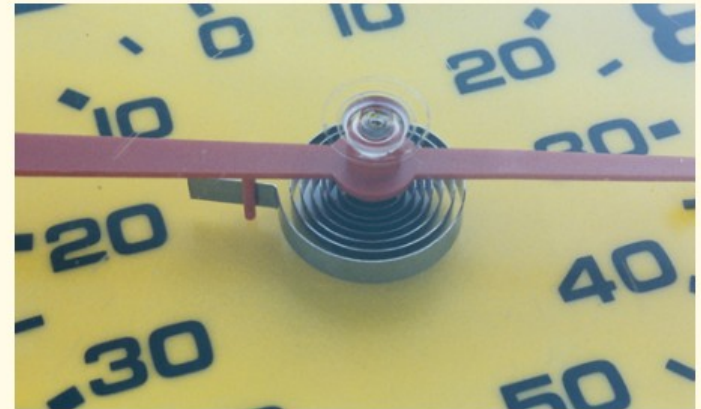


(a)

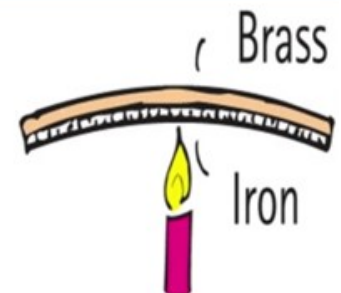
(b) Heated

(c) Cooled

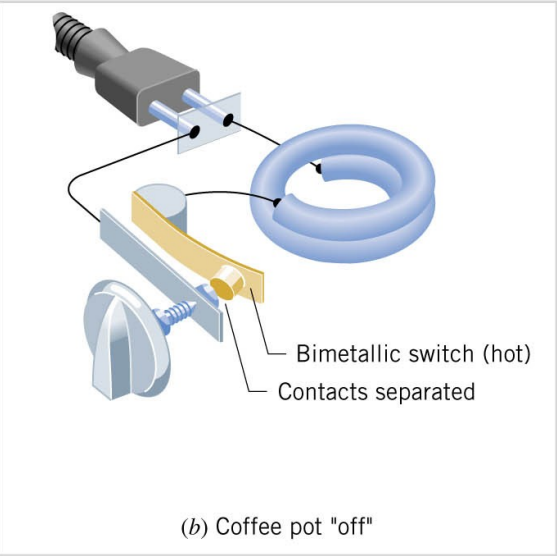
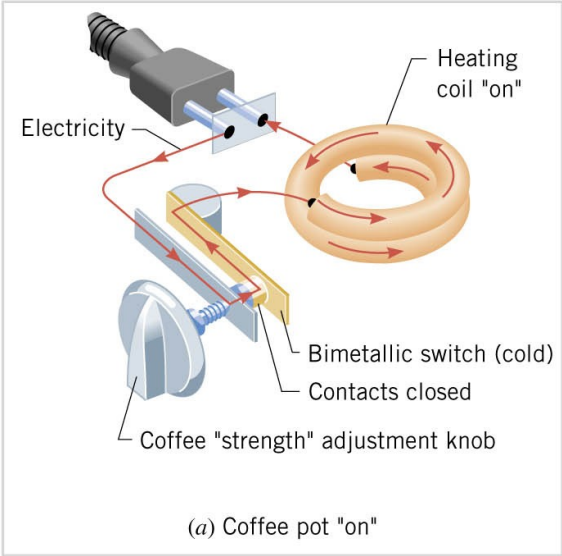
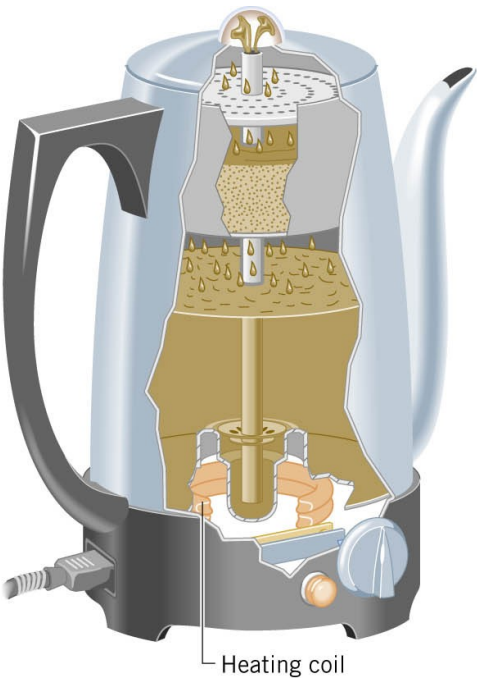
- An analog thermostat is a common example of a bimetallic strip.



Room temperature



12.4 Linear Thermal Expansion



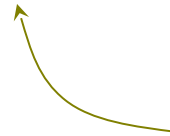


12.5 *Volume Thermal Expansion*

VOLUME THERMAL EXPANSION

The volume of an object changes when its temperature changes:

$$\Delta V = \beta V_o \Delta T$$



coefficient of
volume expansion

Common Unit for the Coefficient of Volume Expansion: $\frac{1}{C^{\circ}} = (C^{\circ})^{-1}$

Material	Type	Expansion coefficient ($^{\circ}\text{C}^{-1}$)
Glass	Length	3.2×10^{-6}
Ice	Length	51×10^{-6}
Steel	Length	10×10^{-6}
Air	Volume	3.7×10^{-3}
Gasoline	Volume	95×10^{-5}
Water, 1°C	Volume	-5×10^{-5}
Water, 20°C	Volume	20×10^{-5}

Expansion of gasoline is
95 parts/ 100,000
For 1 C

So for 20 gallons $\rightarrow$
 $20 \times 95 / 100,000$
For 1 C

And $14 \times 20 \times 95 / 100,000$ for
14 C
That's 0.27 gal/ ($\frac{1}{4}$ about)
That's a lot.

We neglect the change of volume
Of steel

75°F day (24°C)

20 gallon steel gas tank

Fill with gas at 50°F (10°C)

How much gas would get lost?

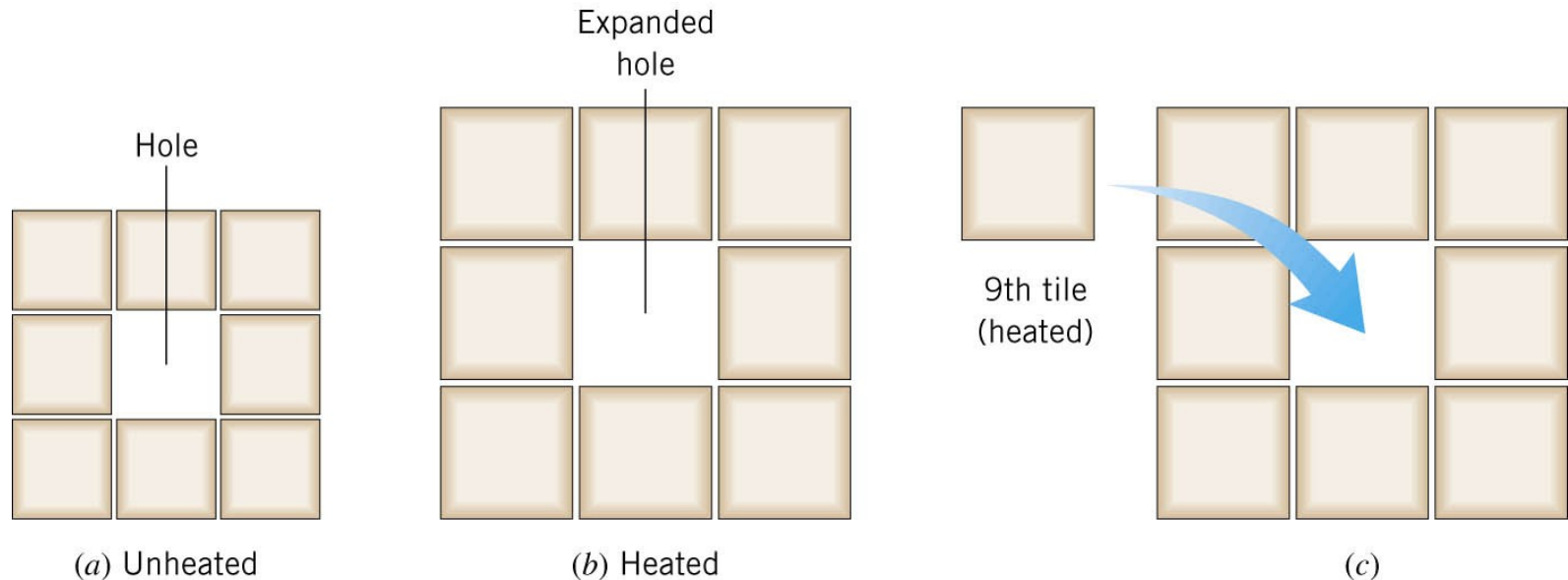
THE EXPANSION OF HOLES

Conceptual Example 5 The Expansion of Holes

The figure shows eight square tiles that are arranged to form a square pattern with a hole in the center. If the tiles are heated, what happens to the size of the hole?

<https://www.youtube.com/watch?v=82FPQ6z8vcE>

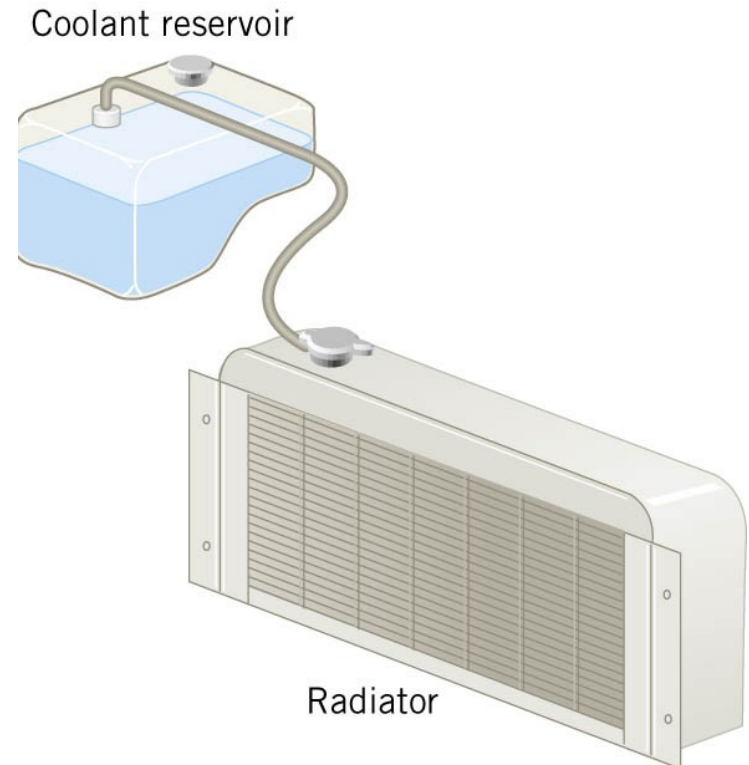
<https://www.youtube.com/watch?v=0PBgFo282IQ&list=PLkRMGCE8BEtZjEVC5J1IRGb9lGT0XOkOJ&index=3>



12.5 Volume Thermal Expansion

Example 8 An Automobile Radiator

A small plastic container, called the coolant reservoir, catches the radiator fluid that overflows when an automobile engine becomes hot. The radiator is made of copper and the coolant has an expansion coefficient of $4.0 \times 10^{-4} (\text{C}^\circ)^{-1}$. If the radiator is filled to its 15-quart capacity when the engine is cold (6°C), how much overflow will spill into the reservoir when the coolant reaches its operating temperature (92°C)?

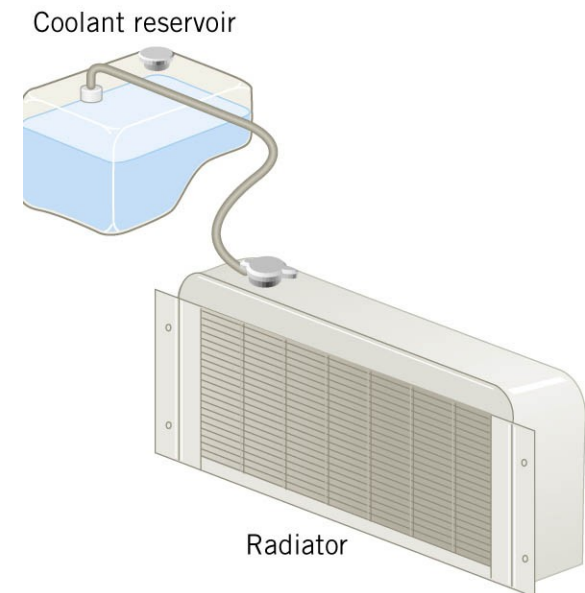


12.5 Volume Thermal Expansion

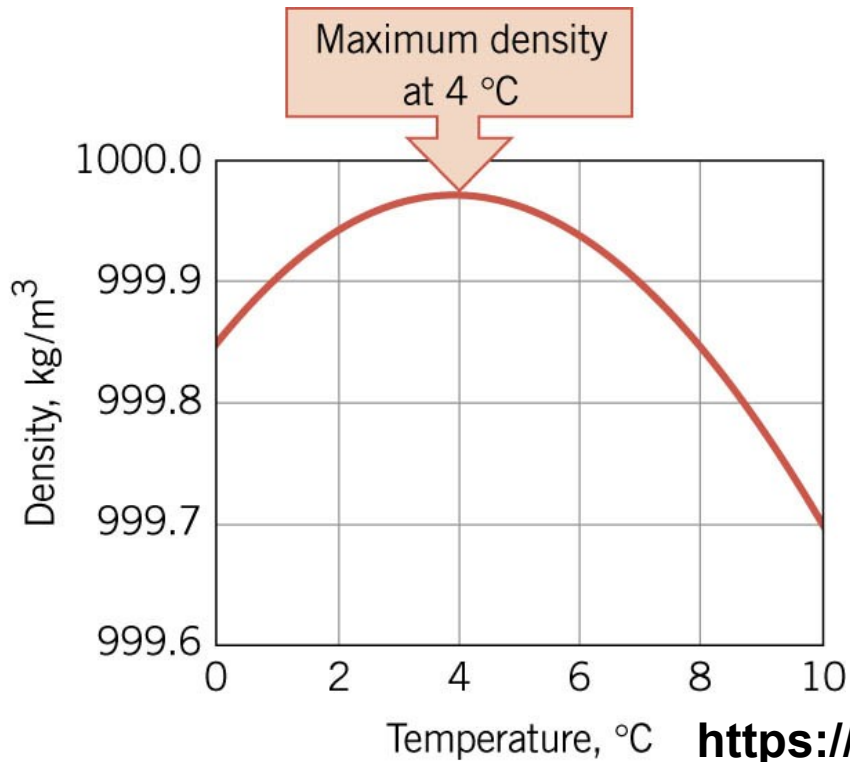
$$\Delta V_{\text{coolant}} = (4.10 \times 10^{-4} (C^\circ)^{-1})(15 \text{ quarts})(86 C^\circ) = 0.53 \text{ quarts}$$

$$\Delta V_{\text{radiator}} = (51 \times 10^{-6} (C^\circ)^{-1})(15 \text{ quarts})(86 C^\circ) = 0.066 \text{ quarts}$$

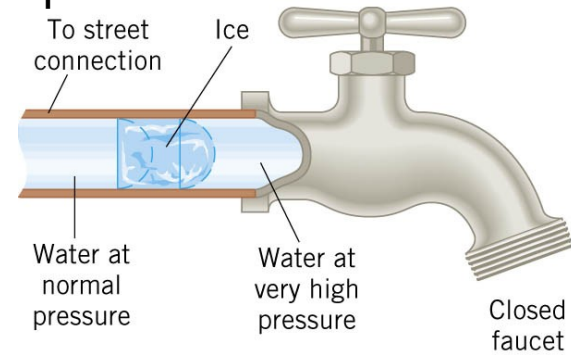
$$\Delta V_{\text{spill}} = 0.53 \text{ quarts} - 0.066 \text{ quarts} = 0.46 \text{ quarts}$$



12.5 Volume Thermal Expansion



Expansion of water.



<https://www.youtube.com/watch?v=u1s17LFXHqo>

